

Nathaniel Sawant

Available for full-time positions starting January 2025

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EDUCATION

Northeastern University, Khoury College of Computer Sciences

Boston, MA

Bachelor of Science in Computer Science, Concentration in Artificial Intelligence

Sep. 2021 – Dec. 2024

GPA: 3.71/4.00

Leadership: Northeastern U. IEEE Chapter (President, *Fall 2023*): NUHOC (VP Publicity, *Summer 2024*)

Relevant Courses: Software Engineering, Database Management Systems, Machine Learning 1 & 2, Object-Oriented Design, Algorithms & Data, C++ Programming, Natural Language Processing, Artificial Intelligence, Prob. & Stats.

Clubs: IEEE (Professional), NUHOC (Outdoors), NUPiC (Photography), Generate (Product Development)

PROFESSIONAL EXPERIENCE

Software Engineer

Jan. 2024 – Jun. 2024

Babel Street, Unstructured Team

Somerville, MA

- Doubled data download speeds with **multi-threading** & cut memory by 20% in Python NER training pipeline.
- Used Python & ontology to analyze NER categorical changes of a couple of million items, ensuring data is stable.
- **Dockerize several Maven projects** and modify existing Jenkins files to increase ease and coverage of testing.

Software Data Engineer

Jan. 2023 – Jun. 2023

Fidelity Investments, Advanced Data Analytics

Boston, MA

- Streamlined development with a **shared code snippet tool** for VSCode/Jupyter Lab using **TypeScript**.
- Conducted **ETL processes** on multiple sites to **scrape data points** to SQL for over 65,000 items with **Python**.
- Setup scrapers using Jenkins, AWS Cloudwatch to store HTML in S3 and formatted data in Oracle SQL databases.

Research Assistant

Oct. 2021 – Aug. 2022

Northeastern University, Rehabilitation Games & Extended Reality Lab

Boston, MA

- Developed two player gameplay mechanics using **Unity 3D engine**, OpenXR, & Photon Unity Networking.
- Captured participants' **eye-tracking data at 60HZ** with Vive Pro to analyze where their attention is focused.

EXTRACURRICULAR EXPERIENCE

Project Lead (DevDB, Snapper)

May. 2024 – Present

Generate Product Development at Northeastern University

Boston, MA

- Led and worked with Agile team to focus **project priorities, deadlines, & deliverables** over 4 month periods.
- **DevDB: Defined system design** for a development database platform using **Node.js, AWS, & PostgreSQL**.
- **Snapper:** Analyzed 5 competitors and worked with design/tech leads & client to plan and develop 6 deliverables.

Software Engineer (Voxeti, Tubender)

Sep. 2023 – Apr. 2024

Generate Product Development at Northeastern University

Boston, MA

- **Voxeti:** Modified C++ open-source 3D printing slicer to estimate print time and deployed as Microservice on EC2.
- **Tubender:** Implemented onboarding flow and settings frontend using **React & Electron** with **C++ backend**.

PROJECTS

Gear Rental Site | *TypeScript, Postgres, Prisma, Next.js, tRPC, NextAuth.js*

Apr. 2024

- Implemented gear rental site for **150+ member** club with gear item checkout/return and **SSO authentication**.
- Designed several pages of **Figma frontend** with shared consistent design (typography, components, palette, etc.)

Reinforcement Learning Agent | *PyTorch, Unity, C#*

Mar. 2023

- Trained and tested **reinforcement learning agent** that seeks target in unknown continuous environment.
- Used **Unity for training and testing** of model in virtual random maze with goal over hundreds of iterations.

TECHNICAL SKILLS

Languages: Python, Java, C#, Golang, C/C++, SQL, Java/TypeScript, HTML/CSS

Frameworks: React, Next.js, Unity, JUnit, FastAPI, Express

Developer Tools: Git, Docker, AWS (S3, EC2, RDS, SageMaker), NoSQL (MongoDB)

Libraries: Pandas, NumPy, Matplotlib, Scikit-learn